SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure C

Case study

Criteria	Understanding of Case ***(2.5)***	Application of Concepts ***(2.5)***	Analysis ***(2)***	Solution Design ***(2)***	Clarity ***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						

Code of Conduct:

I __K Thanuja/192373012__________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
K Thanuja
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A
Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2.Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3.An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)
- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
   - P(Spam), P(Not Spam)
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
   - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

CO3-ATL-2

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out. "wolf"! even though no wolf is in sight. The villagers run to protect the flock, but then get get really mad when they recognize the boy was playing a joke on them. One night, the shepherd boy see a real wolf! The villagers refuse to be fooled again & stay in their houses. The hungry wolf turns the flock into lamb chops. The towns goes hungry. Identify TP, FP, FN, TN.

Act \ Pred	W = yes	W = NO
W=Y (P)	TP Wolf came ✓ warning ✓	FN Wolf came NO warning
W=N (N)	FP Wolf NO came but warning	TN NO wolf came ✓ NO warning ✓

Actual

Prediction

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurence of a cat in 25 of 30 cat images It also predicts absence of cat in 50 of the 70 no cat images. Idenitify TP, FP, FN, TN and calculate the precision & recall and F_1 score

Act \ Pred	cat = Positive	NOcat = Negative
cat = (Positive)	TP = 25 (Actual cat & predicted cat)	FN = 5 (Actual cat but predictive NO cat)
NO cat (Negative)	FP = 20 (Actual NO cat but predicted cat)	TN = 50 Actual NO cat & predicted NO cat

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{25}{25 + 20} = 0.556 \times 100 = (55.6\%)$$

Recall

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$= \frac{25}{25+5}$$

$$= 0.8333 \times 100$$

$$= 83.33\%$$

F1 score

$$F_1 \text{ score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

$$= \frac{2 \times 0.556 \times 83.33}{0.556 + 83.33}$$

$$= 0.6667 \times 100 = 66.67\%$$

3. An email service provider wants to classify emails as spam or not spam using features

contains "offer" (yes/NO)

contains "Free" (yes/NO)

Email length (short/Long)

Sample Dataset

Offer	Free	length	class
yes	yes	short	spam
yes	NO	Long	Not spam
NO	yes	short	spam
NO	NO	Long	Not spam
yes	yes	Long	spam
NO	NO	Short	Not spam

Total = 6

spam = 3

Not spam = 3

$P(\text{spam}) = \frac{3}{6} = \frac{1}{2}$

$P(\text{Not spam}) = \frac{3}{6} = \frac{1}{2}$

2. **Conditional Probability Table**

$P(O = \text{yes}/\text{spam}) = \frac{2}{3}$	$P(O = \text{yes}/\text{No spam}) = \frac{1}{3}$
$P(O = \text{No}/\text{spam}) = \frac{1}{3}$	$P(O = \text{No}/\text{No spam}) = \frac{2}{3}$
$P(F = \text{yes}/\text{spam}) = \frac{3}{3} = 1$	$P(F = \text{yes}/\text{No spam}) = \frac{0}{3}$
$P(F = \text{No}/\text{spam}) = \frac{0}{3}$	$P(F = \text{No}/\text{No spam}) = \frac{3}{3}$
$P(L = \text{yes}/\text{spam}) = \frac{2}{3}$	$P(L = \text{yes}/\text{No spam}) = \frac{1}{3}$
$P(L = \text{No}/\text{spam}) = \frac{1}{3}$	$P(L = \text{No}/\text{No spam}) = \frac{2}{3}$

Using Naive bayes, classify: $P(\frac{A}{B}) = \frac{P(B/A)\,P(A)}{P(B)}$

Offer	Free	Length	Class
yes	yes	short	spam
yes	No	Long	Notspam
No	yes	short	spam
No	No	Long	Notspam
yes	yes	Long	spam
No	No	short	Notspam

For spam $\Rightarrow P(\text{spam}|x) \propto P(\text{spam}) \times P(\text{yes}|\text{spam}) \times P(\text{No}|\text{spam}) \times P(\text{short}|\text{spam})$

$\frac{1}{2} \times \frac{2}{3} \times \frac{0}{3} \times \frac{2}{3} = 0$

For Not spam $\Rightarrow P(\text{Notspam}|x) \propto P(\text{Notspam}) \times P(\text{yes}|\text{Notspam}) \times P(\text{No}|\text{Notspam}) \times P(\text{short}|\text{Notspam})$

$= \frac{1}{2} \times \frac{1}{3} \times \frac{3}{3} \times \frac{1}{3} = \frac{1}{18}$

= 0.056. for Notspam.

∴ P(Notspam|x) > P(spam|x))

Therefore using the navie bayes classifier, the given email(offer = yes, free = NO, Length = short) is classified as NOT spam because its probablity for NOT spam is higher than spam.

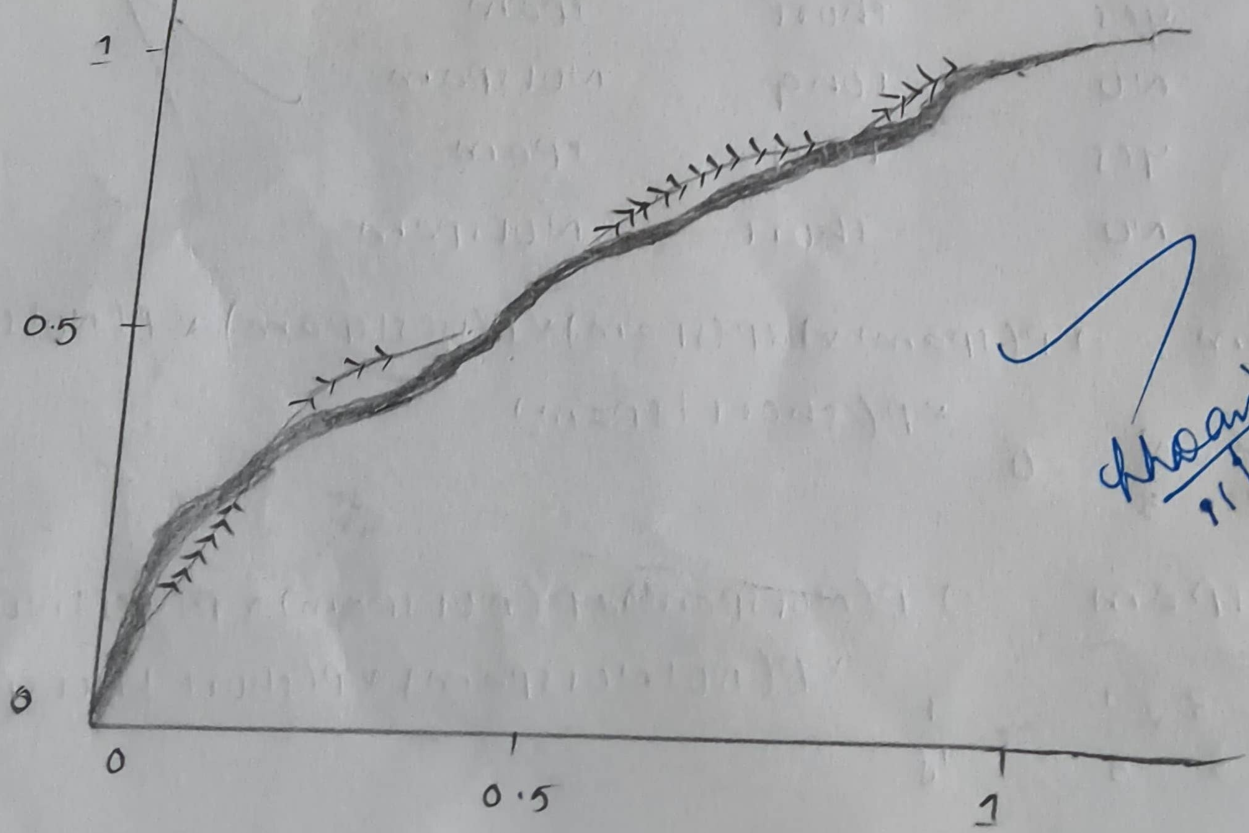
iv) **comparing results using weka**

Dataset	(1) Bayes.Ba	(2) Bayes
Iris	(100) 93.20	95.73
supermarket	(100) 63.71	63.71
	(v/ /*)	(0/2 10)

Key: (i) Bayes.BayesNet
(ii) Bayes. Naive bayes

ROC curve:

x: False Positive rate (Num)
y: True Positive rate (Num)
cold: Threshold (Num)



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